

CANDIDATE'S NAME:.....

SIGNATURE

Index number							



Apex Examination Council

PRE REGISTRATION EXAMINATION SET ONE- 2026

Uganda Certificate of Education

CHEMISTRY

Paper 1(Theory)

TIME: 2 Hours

INSTRUCTIONS TO CANDIDATES;

This paper consists of two sections section **A** and section **B**. It has six (6) items. Section A has two (2) **compulsory** items.

Section B has two parts, part I and part II, Attempt **ONLY** one item from each part.

Answers to section A must be written in the spaces provided while those to section B must be written on the answer booklet(s) provided. Any extra item will not be scored.

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ITEM	1	2	3	4	5	6
SCORE						

SECTION A

Answer all items in this section

ITEM 1

Mzee Larkana a renowned 78 years old herbalist. He is known for his treatment usually from the mixing of a variety of plant parts such as lemon peels, garlic bulbs, gingers and raw honey by getting their extracts, that he uses for treating his trusted patients with digestive related bacterial infections and fu infection among others. However, lately some of his patients he treated didn't fully recover from the diseases instead their health conditions worsened than before and they were admitted in the hospital for proper medication. You have been asked to give guidance.

Task: As a chemistry student:

a) Identify the mistake they made while choosing the products.

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b) Help them understand how the product(s) works.

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c) Suggest the possible side effect(s) of the product to the users and mitigate.

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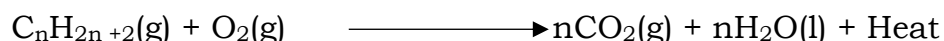
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ITEM 2

The gaseous hydrocarbon X is a carbon based fuel used in racing cars formula 1. Its internal complete engine combustion, gave out carbon dioxide gas, water vapour and heat energy according to the equation below:



The preliminary experiment shows that hydrocarbon X contains 80% carbon and the rest being hydrogen whose relative molecular mass is 30.

However, the ministry of games and sports wants to know how to determine the molecular formula of X and how much volume of the carbon dioxide gas in litres will be emitted if 80g of X were completely burnt to produce heat energy.

Task: Use your
chemistry knowledge
to: a) Explain the:

- i) Category of the homologous series used as carbon – based fuel.

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- ii) Suitability of the gaseous product(s) emitted.

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- b) Guide them on how to determine the:

- (i) Empirical formula and hence molecular formula of X ($C = 12$, $H = 1$, $M_r = 30$)

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- (ii) Volume of carbon dioxide gas emitted into the atmosphere from a single race when hydrocarbon X in (b) (i) is burnt completely in oxygen. (1 mole of a gas occupies 24l at r.t.p; 1mole of X liberated 2moles of carbon dioxide gas)

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d) Advise on the impact arising from using this fuel type to the environment.

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SECTION B- Part I

Answer one item from this part

Item 3

A group of senior 3 students from Gayaza Road Triangle SS went for a study trip to Muko hills in Kabale district, Western Uganda, Where large deposits of iron ores were discovered. The guide disclosed to the students that the government endorsed a Chinese company to extract iron. However, “Residents are worried about how the process is carried out and its environmental implications”. He said.

Task

Make a write up of the presentation you would make to the residents in a sensitization meeting with the Chinese company upon invitation.

Item 4

Health workers have raised concern about rising cases of hypertension (high blood pressure) in hospitals, which they say it is due to continued consumption of cooking oil. Farmers of palm trees in Kalangala are likely to make heavy losses.

Support material



Task

Advise the farmers on how they can benefit from their palm oil by making an alternative product.

Part II

Answer one item from this part

Item 5

An agricultural farm uses diesel-powered machines to till land and transport produces to the market.

Support material



The government officials have invited students' representatives from your school to farm to educate the workers about the fuel used.

Task

Write the information you would deliver to the workers as the key note speaker.

Item 6

Namanve forest reserve was cleared to set up an industrial park, harboring factories. This heavily altered the quality of air in the area due to waste fumes from factories and engines of heavy cargo trucks.

The District Environment Officer, Wakiso district has organized a meeting to create awareness to the industry operators

Support material



Task

As a learner of chemistry, write your message to the industry operators on behalf of the District Environment Officer.

END



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